XRPL Validator — Daily Engineering Report · 2026-10-15

Period: 2026-10-15 00:00 to 24:00 US Eastern (2026-10-15 04:00 to 2026-10-16 04:00 UTC); ledgers #107,664,778 to #107,687,072. Every count from the ledger and the validator's logs stops at #107,687,072, closed 03:59:51 UTC. Generated 2026-10-21 00:48 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records; Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores. Not available: none.

Status

Impaired (all causes resolved or fixed) — 4 validations of a ledger hash the network did not validate.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,267 of 22,295 ledgers; 4 validations did NOT match the network's validated ledger. Unsigned: 28 during planned restarts.
- **Native engine:** 0 engine receipts recorded.
- **Changes:** 3 deployments (cycles 177, 178, 179), 12 commits merged into main.
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-10-23 09:34 UTC.

Service levels

Measure	Value
Ledgers closed by the network	22,295
Validations issued	22,267
Coverage (validations / ledgers closed)	99.874%
Validations matching the network's validated ledger hash	22,263 of 22,267
Unsigned during planned restarts	28
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%
Restarts	3

Measure	Value
Basis: the hash the first peer announced as accepted (before cycle 183)	22,267
External agreement, data.xrpl.org (UTC day 2026-10-15)	0.99803 (42 missed of 21,366)

Restarts:

- 14:12:19 UTC: deployment of cycle 177; validations resumed at #107,674,267 (10 ledgers unsigned between the last validation before the restart and the first after it)
- 15:43:36 UTC: deployment of cycle 178; validations resumed at #107,675,675 (9 ledgers unsigned between the last validation before the restart and the first after it)
- 16:29:40 UTC: deployment of cycle 179; validations resumed at #107,676,389 (9 ledgers unsigned between the last validation before the restart and the first after it)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,664,778–#107,687,072), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipts this day.

Incidents

INC-01 · 4 validations of a ledger hash the network did not validate

Window	2026-10-15 04:08 UTC to 2026-10-16 01:53 UTC
Ledgers	#107,664,904, #107,670,768, #107,672,774, #107,685,116
Impact	the validator signed a hash the network did not validate for these ledgers; data.xrpl.org scores such ledgers missed

Cause	Before cycle 183 the validator signed the ledger hash from the first peer that announced a ledger as accepted, without confirmation. For these ledgers that peer had built a different ledger, so the validator signed a hash the network never validated.
Action	commit d540e6e — fix(validator): sign a ledger only once the trusted validators reach quorum on its hash; deployed in cycle 183 at 2026-10-18 22:18 UTC

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
177	`0c32cdf`	14:13:08 UTC	2,120 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	407 matched, 0 mismatched	#52 from 14:39 UTC: ended early: superseded by the next deployment (2026-10-15 15:44 UTC)
178	`d7f5d8b`	15:44:25 UTC	2,139 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	413 matched, 0 mismatched	#53 from 16:11 UTC: ended early: superseded by the next deployment (2026-10-15 16:30 UTC)
179	`046efb7`	16:30:30 UTC	2,156 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	407 matched, 0 mismatched	#54 from 16:56 UTC: ended early: superseded by the next deployment (2026-10-16 16:59 UTC)

Cycle 177 delivered: `6d81d3e` feat(peer): Independence 7.1 — answer regular peers' pings (XRPL_PEER_PONG=1, off by default); the plan gains steps 7-9; `0c32cdf` fix(peer): 7.1's pong is written from its own task, so a GetLedger reply holding the writer through a long write to a slow peer never stalls the read loop (a5's review)

Cycle 178 delivered: `5a2c738` feat(peer): Independence 7.2-7.4 — one node key, private by design, checked transaction relay (all off by default); `c8c361d` feat(validator): Independence Step 9 — clock check, pending-only amendment votes, the list refreshed while running (all off by default); `d7f5d8b` fix(unl): drop the unused read_vl_length import (clippy, a5's o183)

Cycle 179 delivered: `58158db` feat(peer-state): Independence 8 in shadow — a ledger's whole state tree from mainnet peers, every node proven (peer_state_sync); `79e684c` feat(peer-state): per-peer reply sizes, latency, timeouts and close reasons at the end; discovery in the progress line; `88df3b3` feat(peer-state): the plan records Step 8 built in shadow, with the measured full run; `046efb7` fix(peer-state): depth.is_multiple_of(2) (clippy, a5's o184)

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
13:32 UTC	xrpl-validator	`6d81d3e`	feat(peer): Independence 7.1 — answer regular peers' pings (XRPL_PEER_PONG=1, off by default); the plan gains steps 7-9
13:32 UTC	xrpl-validator	`0c32cdf`	fix(peer): 7.1's pong is written from its own task, so a GetLedger reply holding the writer through a long write to a slow peer never stalls the read loop (a5's review)
14:13 UTC	xrpl-validator	`5a2c738`	feat(peer): Independence 7.2-7.4 — one node key, private by design, checked transaction relay (all off by default)
14:13 UTC	xrpl-validator	`c8c361d`	feat(validator): Independence Step 9 — clock check, pending-only amendment votes, the list refreshed while running (all off by default)

Merged	Repo	Commit	Subject
14:13 UTC	xrpl-validator	`d7f5d8b`	fix(unl): drop the unused read_vl_length import (clippy, a5's o183)
15:45 UTC	xrpl-validator	`88df3b3`	feat(peer-state): the plan records Step 8 built in shadow, with the measured full run
15:45 UTC	xrpl-validator	`79e684c`	feat(peer-state): per-peer reply sizes, latency, timeouts and close reasons at the end; discovery in the progress line
15:45 UTC	xrpl-validator	`58158db`	feat(peer-state): Independence 8 in shadow — a ledger's whole state tree from mainnet peers, every node proven (peer_state_sync)
15:45 UTC	xrpl-validator	`046efb7`	fix(peer-state): depth.is_multiple_of(2) (clippy, a5's o184)
18:54 UTC	xrpl-validator	`1711cca`	chore(tools): xrpl-watch moved to its own repo (ilubxrp589/xrpl-watch); this folder keeps a pointer
20:22 UTC	xrpl-validator	`0bb7783`	chore: scrub a login password and home-network addresses from docs, the dashboard config and a script
20:39 UTC	xrpl-validator	`67ce8ff`	chore(node): the source names no home-network address

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **feat(peer): Independence 7.1 — answer regular peers' pings (XRPL_PEER_PONG=1, off by default); the plan gains steps 7-9** (`6d81d3e` , xrpl-validator) Regular peer sessions never answered the 60-second keep-alive ping, so every server dropped the validator host after about two minutes and it dialled again: about 25,000 handshakes in the first 10 hours of cycle 176, for some 80 peers. Only the 4 relay sessions answered (peer::relay). With XRPL_PEER_PONG=1 every session answers with peer::codec::pong_frame_for (tested in codec.rs), before the duplicate filter. /api/peers gains …
- **fix(peer): 7.1's pong is written from its own task, so a GetLedger reply holding the writer through a long write to a slow peer never stalls the read loop (a5's review)** (`0c32cdf` , xrpl-validator)
- **feat(peer): Independence 7.2-7.4 — one node key, private by design, checked transaction relay (all off by default)** (`5a2c738` , xrpl-validator) 7.2 XRPL_NODE_IDENTITY=1: every regular peer session shows one node key, loaded from paths::node_seed_path (XRPL_NODE_SEED_PATH, default ${XRPL_DATA_DIR}/node_seed.hex, 0600, created on first use by load_or_create_seed), never the validator's seed (refused if the paths match). Without it each connection made a fresh key, so every server met a stranger each time the validator host dialled it. A seed that can't be used falls back …
- **feat(validator): Independence Step 9 — clock check, pending-only amendment votes, the list refreshed while running (all off by default)** (`c8c361d` , xrpl-validator) Clock (XRPL_CLOCK_CHECK=1, XRPL_CLOCK_GATE=1): clock_check.rs, a small SNTP client (RFC 4330, no new dependency) that asks time.google.com, time.cloudflare.com and pool.ntp.org every 5 minutes and keeps the median offset: a log line when it changes, `clock` in /api/engine. rippled 3.x has no such check (TimeKeeper::now() is the system clock, TimeKeeper.h:66). The gate skips signing while 2 or more servers agree the …
- **fix(unl): drop the unused read_vl_length import (clippy, a5's o183)** (`d7f5d8b` , xrpl-validator)

- **feat(peer-state): the plan records Step 8 built in shadow, with the measured full run** (`88df3b3`, xrpl-validator) docs/INDEPENDENCE-PLAN.md, Step 8: on a build host, ledger #107674282 (a 28/28 quorum of the verified list), all 20,031,916 objects from 12 mainnet peers in 14 minutes (5,436 requests, 7.0 GB in, 0.2 GB of memory, no timeout, refusal or disconnect); the root rebuilt bottom-up from the file equals the account_hash, and the reference node agreed as a witness. What the source switch still needs: writing straight into state.rocks and using …
- **feat(peer-state): per-peer reply sizes, latency, timeouts and close reasons at the end; discovery in the progress line** (`79e684c`, xrpl-validator) What each peer allowed is the point of a shadow run: the biggest reply (nodes and MB), the average time to answer, the timeouts and why each session closed, printed per peer after the run, and the addresses TMEndpoints brought in the progress line and the summary.
- **feat(peer-state): Independence 8 in shadow — a ledger's whole state tree from mainnet peers, every node proven (peer_state_sync)** (`58158db`, xrpl-validator) A cold start downloads the state from the reference node's RPC (ledger_data, bulk_sync.rs); nothing fetched it from peers. peer_state does what rippled's InboundLedger does: TMGetLedger liAS_NODE by position, each node taken only where a proven parent named it and only when it hashes to that name (inner: MIN over the 16 children; leaf: MLN over the object and its key, the key under its position), starting from the header's …
- **fix(peer-state): depth.is_multiple_of(2) (clippy, a5's o184)** (`046efb7`, xrpl-validator)

Docs and operations

- **chore(tools): xrpl-watch moved to its own repo (ilubxrp589/xrpl-watch); this folder keeps a pointer** (`1711cca`, xrpl-validator)
- **chore: scrub a login password and home-network addresses from docs, the dashboard config and a script** (`0bb7783`, xrpl-validator)
- **chore(node): the source names no home-network address** (`67ce8ff`, xrpl-validator)

Independence program

No independence record for UTC day 2026-10-15; records begin 2026-10-20.

Network

Measure	Value
Ledgers closed	22,295 (#107,664,778–#107,687,072)
Mean close interval	3.88 s
Longest close interval	10 s (before #107,686,255)
Closes at least 15 s apart	0
Empty ledgers	0

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-09 09:34 UTC	2026-10-23 09:34 UTC

Amendments enabled on mainnet: 97.

Open items

- Amendment fixCleanup3_4_0: activation due ~2026-10-23 09:34 UTC.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
10-15 13:30	10-15 13:32	xrpl-validator	`6d81d3e`	feat(peer): Independence 7.1 — answer regular peers' pings (XRPL_PEER_PONG=1, off by default); the plan gains steps 7-9
10-15 13:31	10-15 13:32	xrpl-validator	`0c32cdf`	fix(peer): 7.1's pong is written from its own task, so a GetLedger reply holding the writer through a long write to a slow peer never stalls the read loop (a5's review)
10-15 13:49	10-15 14:13	xrpl-validator	`5a2c738`	feat(peer): Independence 7.2-7.4 — one node key, private by design, checked transaction relay (all off by default)
10-15 14:03	10-15 14:13	xrpl-validator	`c8c361d`	feat(validator): Independence Step 9 — clock check, pending-only amendment votes, the list refreshed while running (all off by default)
10-15 14:05	10-15 14:13	xrpl-validator	`d7f5d8b`	fix(unl): drop the unused read_vl_length import (clippy, a5's o183)
10-15 14:29	10-15 15:45	xrpl-validator	`88df3b3`	feat(peer-state): the plan records Step 8 built in shadow, with the measured full run
10-15 14:29	10-15 15:45	xrpl-validator	`79e684c`	feat(peer-state): per-peer reply sizes, latency, timeouts and close reasons at the end; discovery in the progress line
10-15 14:29	10-15 15:45	xrpl-validator	`58158db`	feat(peer-state): Independence 8 in shadow — a ledger's whole state tree from mainnet peers, every node proven (peer_state_sync)
10-15 15:45	10-15 15:45	xrpl-validator	`046efb7`	fix(peer-state): depth.is_multiple_of(2) (clippy, a5's o184)
10-15 18:54	10-15 18:54	xrpl-validator	`1711cca`	chore(tools): xrpl-watch moved to its own repo (ilubxrp589/xrpl-watch); this folder keeps a pointer
10-15 20:22	10-15 20:22	xrpl-validator	`0bb7783`	chore: scrub a login password and home-network addresses from docs, the dashboard config and a script
10-15 20:36	10-15 20:39	xrpl-validator	`67ce8ff`	chore(node): the source names no home-network address

Times in UTC. A commit appears on the day it was committed to main or merged into it.